The Indian Monsoon

Seasons of wind

The word monsoon comes from the Arabic word mausim, meaning season. It describes winds that reverse their direction with the seasons. In summer the land of the Indian subcontinent heats up much faster than the Indian Ocean. Hot air over the land rises and creates low pressure, while the cooler ocean has high pressure. Moist winds blow from the sea towards the land.

The south-west monsoon

The south-west monsoon reaches Kerala around the first of June. It splits into two branches: the Arabian Sea branch, which strikes the Western Ghats, and the Bay of Bengal branch, which moves up the north-east and then turns west along the Himalayas. By mid-July the monsoon covers the whole country.

Mountains decide where the rain falls. Mawsynram in Meghalaya receives about 11,870 millimetres of rain in a year, among the highest in the world, while the Thar desert in Rajasthan gets less than 150 millimetres. The windward side of the Western Ghats is wet; the leeward side, in the rain shadow, is dry.

The retreating monsoon

In October and November the land cools, the low pressure moves south and the winds reverse, blowing from land to sea. This is the retreating, or north-east, monsoon. It brings rain to Tamil Nadu, which gets most of its rainfall in these months.

Why it matters

About half of India's farmland depends on rain, not irrigation. Crops sown with the monsoon, such as rice, cotton and soybean, are called kharif crops. A weak monsoon can mean drought, lower farm incomes and higher food prices; a very heavy one can bring floods in Assam and Bihar. El Nino, a warming of the Pacific Ocean, often weakens the Indian monsoon.